Constant rates of change

1

a Constant b Variable c Constant d Variable
e Variable f Variable

2 Constant

Instantaneous rates of change

3 0

4 0

5 2

6 2

7

a 0 b 0 c 2 d 1
e -2 f 2 g -1 h 1
i -3

8 7

9 -6.5

10 2

11 -2

12 $\frac{1}{2}$

13 2

14 -2

a $a = 7$ **b** $b = 5$ **c** 2

16

a

a	b	$h = b - a$	$\dfrac{f(b) - f(a)}{b - a}$
1	2	1	7
1	1.5	0.5	4.75
1	1.1	0.1	3.31
1	1.05	0.05	3.1525
1	1.01	0.01	3.0301
1	1.001	0.001	3.003
1	1.0001	0.0001	3.0003

b 3

17

a

a	b	$h = b - a$	$\dfrac{f(b) - f(a)}{b - a}$
1	2	1	6
1	1.5	0.5	5
1	1.1	0.1	4.2
1	1.05	0.05	4.1
1	1.01	0.01	4.02
1	1.001	0.001	4.002
1	1.0001	0.0001	4.0002

b 4

18

a

a	b	$h = b - a$	$\dfrac{f(b) - f(a)}{b - a}$
0	1	1	3
0	0.5	0.5	2
0	0.1	0.1	1.487
0	0.05	0.05	1.4355
0	0.01	0.01	1.3959
0	0.001	0.001	0.138 73
0	0.0001	0.0001	1.3864

b 1.386

19

a

a	b	$h = b - a$	$\dfrac{f(b) - f(a)}{b - a}$
1	2	1	-3
1	1.5	0.5	-2.5
1	1.1	0.1	-2.1
1	1.05	0.05	-2.05
1	1.01	0.01	-2.01
1	1.001	0.001	-2.001
1	1.0001	0.0001	-2.0001

b -2

20

a

a	b	$h=b-a$	$\frac{f(b)-f}{b-a}$ [redacted]
1	2	1	-2.5
1	1.5	0.5	-3.3333
1	1.1	0.1	-4.5455
1	1.05	0.05	-4.7619
1	1.01	0.01	-4.9505
1	1.001	0.001	-4.995
1	1.0001	0.0001	-4.9995

b -5

Applications

21 $-\$10$ per $1^\circ F$

22

a

Year	0	1	2	3	4	5
Tobia's staff	310	269	228	187	146	105
Gwen's staff	310	302	286	262	230	190

b -41

c -32

d Linear, because it decreases by the same amount each year.

e Non-linear, because it decreases by more each year.

f Tobias's

g Gwen's

23

a

Month	0	1	2	3	4	5
Mario's clients	0	27	54	81	108	135
Christa's clients	0	6	18	36	60	90

b Linear

- **d** 27
- **e** 18
- **f** Mario's
- **f** Christa's

24

a

Month	0	1	2	3	4	5
Value of Harry's investment	2000	2060	2120	2180	2240	2300
Value of Skye's investment	2000	2040	2080.8	2122.42	2164.87	2208.17

b $\$60$

c $\$40.8$

d Linear

e Non-linear

f Harry's

25

a $-\frac{1}{2}$

b The amount of water in the bucket decreases by $\frac{1}{2}$ litre every minute.

26

a $\frac{3}{5}$

b The cost of a call increases by $\$0.60$ for each additional minute

27 Oliver, 10 km/h

28

a 0°F per hour **b** 6°F per hour **c** -1°F per hour **d** 1.5°F per hour

29

a

i 9 m/s **ii** 3 m/s **iii** -3m/s **iv** -9 m/s

b Between $t = 0$ and $t = 1$, and between $t = 3$ and $t = 4$

f The ball is falling down.